

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<think> I need to consider the location of the wound and the anatomy of the hand to determine which artery is most likely damaged. The 'anatomical snuffbox' is a location on the radial, dorsal aspect of the wrist, where the radial artery is closely related to the scaphoid bone. Given the location of the wound and the symptoms of excessive bleeding, it's likely that the radial artery is compromised. The ulnar artery is located more posteriorly and on the ulnar side of the wrist, and the Princeps pollicis artery is a small branch of the radial artery that supplies the thumb. The brachial artery is a larger artery that arises from the axilla and runs down the arm to supply the elbow and forearm. Considering the anatomical layout, the radial artery is most likely to be responsible for the bleeding due to the location of the wound in the anatomical snuffbox. </think><answer> A. Radial artery </answer>

